

sed Cheat Sheet

Edit text line by line: find and replace, delete or print lines, insert text.

Using it

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sed [options] 'script' file ← prints the edited text; the file is NOT changed
sed -i 'script' file ← edits the file itself
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Option	What it does	Option	What it does
-n	Prints only what you ask for (with p)	-E	Extended patterns (+ ? ())
-i	Edits the file in place	-e	Adds another script (repeatable)
-i.bak	Edits in place, keeping a .bak backup	-f file	Reads the script from a file

Find and replace

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s/old/new/flags ← replace old (a pattern) with new
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Piece	Meaning	Piece	Meaning
g	Replaces every match on the line (default: first only)	&	In the replacement: the whole matched text
2	Replaces only the 2nd match on each line	\1 \2	In the replacement: groups () from the pattern
I	Ignores upper/lower case (GNU sed)	\/	A literal / inside the pattern
p	Prints the line if it changed (use with -n)	s a b 	Any character can be the separator

Choosing lines (addresses)

Address	Lines it picks	Address	Lines it picks
5	Line 5	/error/	Lines containing error
5,10	Lines 5 to 10	/start/,/end/	From a line matching start to one matching end
\$	The last line	1~2	Every 2nd line from line 1 (GNU sed)
5,\$	Line 5 to the end	/error/!	! flips it: lines not containing error

Put the address in front of a command: **5,10d** deletes lines 5–10; **/error/s/a/b/** replaces only on lines containing error. No address = every line.

Commands

Command	Does	Command	Does
s	Substitute (find and replace)	i\ text	Inserts a line before
d	Deletes the line	a\ text	Appends a line after
p	Prints the line	c\ text	Replaces the whole line
q	Quits (stops reading)	=	Prints the line number

Handy recipes

Command	Does
sed 's/foo/bar/g' file	Shows the file with every foo replaced by bar
sed -i.bak 's/8080/9090/' app.conf	Changes the file itself, keeping app.conf.bak
sed -n '10,20p' file	Prints only lines 10–20
sed '/^#/d; /^\$/d' file	Removes comment lines and blank lines
sed 's/[:space:]]*\$// ' file	Removes trailing spaces and tabs
sed '1d' data.csv	Drops the header line
sed -E 's/(\w+) (\w+)/\2 \1/' file	Swaps the first two words on each line
sed 's /usr/local /opt g' file	Replaces paths without escaping every /

Remember: run it **without -i** first and check the output. On macOS, **-i** needs a suffix argument: **sed -i '' 's/a/b/' file**.